

Section-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Find equation of parabola whose vertex is (0,0) passing through (5, 2) and symmetric with respect to y-axis. (CO4)

Q.20 Find the 3rd term from the end in the expression of $\left(2x + \frac{1}{2x}\right)^{10}$. (CO3)

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**Level 3, 1st Sem. / DVOC (Ref. & Air Cond.,
Auto Servicing, ITM, PT, SD, AMT, FP, EMS)**

Subject : Applied Mathematics-I

Time : 2 Hrs.

M.M. : 50

Section-A

Note: Multiple Choice questions. Attempt all questions. (5x1=5)

Q.1 The roster form of the set $\{x: x \text{ is a whole number and } -1 < x < 3\}$ is (CO1)

- (a) $\{-1, 0, 1, 2, 3, 4\}$ (b) $\{1, 2, 3\}$
(c) $\{0, 1, 2, 3\}$ (d) None of these

Q.2 The missing term in the GP 2, 6, 18, ____, 162 is (CO2)

- (a) 54 (b) 36
(c) 9 (d) None of these

Q.3 What is the real part of the complex no $z = 5i - 2$ is (CO3)

- (a) 5 (b) 2
(c) -5 (d) -2

Q.4 The slope of straight line making an angle of 30° with positive x -axis is _____ (CO4)

- (a) 0 (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $\frac{1}{3}$

Q.5 If one coin is tossed two times then find the probability of getting both heads. (CO5)

- (a) 1 (b) 0
(c) $\frac{1}{4}$ (d) $\frac{1}{3}$

Section-B

Note: Very short answer questions. Attempt all questions. (5x1=5)

Q.6 The value of $\cos 45^\circ$ is _____. (CO1)

Q.7 $3+6+9+12+\dots$ is A.P. (True/False) (CO2)

Q.8 $\frac{6!}{4!} =$ _____. (CO3)

Q.9 The slope of line which makes an angle of 60° with positive x -axis is _____. (CO4)

Q.10 If an event has more than one sample point, then it is called simple event. (True/False) (CO5)

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Section-C

Note: Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Let $f(x)=x^2$ and $g(x)=2x+1$, then find the following: (CO1)

$$(f+g)(x), (f-g)(x), (fg)(x) \text{ and } \left(\frac{f}{g}\right)(x).$$

Q.12 In a GP the 3rd term is 24 and 6th term is 192. Find its 9th term. (CO2)

Q.13 Express $z = \frac{1}{2+3i}$ in standard form $x+iy$. (CO3)

Q.14 Find equation of circle whose end points of diameter are $(-1, 2)$ and $(3, 4)$. (CO4)

Q.15 Two unbiased coins are tossed simultaneously find probability of getting one head. (CO5)

Q.16 Let $f(x)=\sqrt{x}$ and $g(x)=x$, then find the following: (CO1)

$$(f+g)(x), (f-g)(x), (fg)(x) \text{ and } \left(\frac{f}{g}\right)(x).$$

Q.17 Find 3rd term from the end of the Binomial expression of $(x+2y)^{10}$. (CO3)

Q.18 Find the sum n terms of the series whose n^{th} term is $(n+1)^2$. (CO2)

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